Newman-Penrose Formalism Calculations

Stephani et al., Equation (31.57)

Metric

$$ds^2 = (-2\,v^2(\frac{\partial\,f_x}{\partial x})^2 - 4\,v^2\frac{\partial\,f_x}{\partial x}\frac{\partial\,f_x}{\partial y} - 2\,v^2\frac{\partial\,f_x}{\partial y}^2 - 2\,v^2(\frac{\partial\,f_y}{\partial x})^2 - 4\,v^2\frac{\partial\,f_y}{\partial x}\frac{\partial\,f_y}{\partial y} - 2\,v^2\frac{\partial\,f_y}{\partial y}^2 - 2\,H^0(u,v))\mathrm{d}u\otimes\mathrm{d}u - \mathrm{d}u \otimes \mathrm{d}v - \mathrm{d}v\otimes\mathrm{d}u + (\frac{1}{f_x^4 + f_y^4 + 2\,(f_x^2+1)f_y^2 + 2\,f_x^2 + 1})\mathrm{d}x\otimes\mathrm{d}x + (\frac{1}{f_x^4 + f_y^4 + 2\,(f_x^2+1)f_y^2 + 2\,f_x^2 + 1})\mathrm{d}y\otimes\mathrm{d}y$$

Coordinates

$(x^\mu) = (u, v, x, y)$

Metric Functions

$f_x(x,y) = f_x$

$f_y(x,y) = f_y$

Null Tetrad

Covariant components

$l = (1)\,du$

$$n = (v^2\frac{\partial}{\partial x}f_x^2 + 2\,v^2\frac{\partial}{\partial x}f_x\frac{\partial}{\partial y}f_x + v^2\frac{\partial}{\partial y}f_x^2 + v^2\frac{\partial}{\partial x}f_y^2 + 2\,v^2\frac{\partial}{\partial x}f_y\frac{\partial}{\partial y}f_y + v^2\frac{\partial}{\partial y}f_y^2 + H^0(u,v))\,du + (1)\,dv$$

$$m = (\frac{\sqrt{\frac{1}{2}}}{\sqrt{f_x^4 + 2\,f_x^2 f_y^2 + f_y^4 + 2\,f_x^2 + 2\,f_y^2 + 1}})\,dx + (\frac{i\,\sqrt{\frac{1}{2}}}{\sqrt{f_x^4 + 2\,f_x^2 f_y^2 + f_y^4 + 2\,f_x^2 + 2\,f_y^2 + 1}})\,dy$$

$$\overline{m} = (\frac{\sqrt{\frac{1}{2}}}{\sqrt{f_x^4 + 2\,f_x^2 f_y^2 + f_y^4 + 2\,f_x^2 + 2\,f_y^2 + 1}})\,dx + (-\frac{i\,\sqrt{\frac{1}{2}}}{\sqrt{f_x^4 + 2\,f_x^2 f_y^2 + f_y^4 + 2\,f_x^2 + 2\,f_y^2 + 1}})\,dy$$

Contravariant components

$$l = (-1)\,dv$$

$$n = (-1)\,du + (v^2 \frac{\partial}{\partial x} f_x^2 + 2\,v^2 \frac{\partial}{\partial x} f_x \frac{\partial}{\partial y} f_x + v^2 \frac{\partial}{\partial y} f_x^2 + v^2 \frac{\partial}{\partial x} f_y^2 + 2\,v^2 \frac{\partial}{\partial x} f_y \frac{\partial}{\partial y} f_y + v^2 \frac{\partial}{\partial y} f_y^2 + H^0(u,v))\,dv$$

$$m = (\frac{1}{2}\,\sqrt{2} f_x^2 + \frac{1}{2}\,\sqrt{2} f_y^2 + \frac{1}{2}\,\sqrt{2})\,dx + (\frac{1}{2} i\,\sqrt{2} f_x^2 + \frac{1}{2} i\,\sqrt{2} f_y^2 + \frac{1}{2} i\,\sqrt{2})\,dy$$

$$\overline{m} = (\frac{1}{2}\,\sqrt{2} f_x^2 + \frac{1}{2}\,\sqrt{2} f_y^2 + \frac{1}{2}\,\sqrt{2})\,dx + (-\frac{1}{2} i\,\sqrt{2} f_x^2 - \frac{1}{2} i\,\sqrt{2} f_y^2 - \frac{1}{2} i\,\sqrt{2})\,dy$$

Directional Derivatives

$$D = -\partial_v$$

$$\Delta = -\partial_u + v^2 \frac{\partial}{\partial x} f_x^2 + 2\,v^2 \frac{\partial}{\partial x} f_x \frac{\partial}{\partial y} f_x + v^2 \frac{\partial}{\partial y} f_x^2 + v^2 \frac{\partial}{\partial x} f_y^2 + 2\,v^2 \frac{\partial}{\partial x} f_y \frac{\partial}{\partial y} f_y + v^2 \frac{\partial}{\partial y} f_y^2 + H^0(u,v)\,\partial_v$$

$$\delta = \frac{1}{2}\,\sqrt{2} f_x^2 + \frac{1}{2}\,\sqrt{2} f_y^2 + \frac{1}{2}\,\sqrt{2}\,\partial_x + \frac{1}{2} i\,\sqrt{2} f_x^2 + \frac{1}{2} i\,\sqrt{2} f_y^2 + \frac{1}{2} i\,\sqrt{2}\,\partial_y$$

$$\overline{\delta} = \frac{1}{2}\,\sqrt{2} f_x^2 + \frac{1}{2}\,\sqrt{2} f_y^2 + \frac{1}{2}\,\sqrt{2}\,\partial_x - \frac{1}{2} i\,\sqrt{2} f_x^2 - \frac{1}{2} i\,\sqrt{2} f_y^2 - \frac{1}{2} i\,\sqrt{2}\,\partial_y$$

Spin Coefficients

$$\rho = 0$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = 0$$

$$\lambda = 0$$

$$\begin{aligned}
\nu = {} & \sqrt{2}v^2 f_x^2 \frac{\partial}{\partial x} f_x \frac{\partial^2}{(\partial x)^2} f_x + \sqrt{2}v^2 f_y^2 \frac{\partial}{\partial x} f_x \frac{\partial^2}{(\partial x)^2} f_x - (i-1)\sqrt{2}v^2 f_x^2 \frac{\partial}{\partial x} f_x \frac{\partial^2}{\partial x \partial y} f_x - (i-1)\sqrt{2}v^2 f_y^2 \frac{\partial}{\partial x} f_x \frac{\partial^2}{\partial x \partial y} f_x + \sqrt{2}v^2 f_x^2 \frac{\partial^2}{(\partial x)^2} f_x \frac{\partial}{\partial y} f_x + \sqrt{2}v^2 f_y^2 \frac{\partial^2}{(\partial x)^2} f_x \frac{\partial}{\partial y} f_x \\
& - (i-1)\sqrt{2}v^2 f_x^2 \frac{\partial^2}{\partial x \partial y} f_x \frac{\partial}{\partial y} f_x - (i-1)\sqrt{2}v^2 f_y^2 \frac{\partial^2}{\partial x \partial y} f_x \frac{\partial}{\partial y} f_x - i\sqrt{2}v^2 f_x^2 \frac{\partial}{\partial x} f_x \frac{\partial^2}{(\partial y)^2} f_x - i\sqrt{2}v^2 f_y^2 \frac{\partial}{\partial x} f_x \frac{\partial^2}{(\partial y)^2} f_x - i\sqrt{2}v^2 f_x^2 \frac{\partial}{\partial y} f_x \frac{\partial^2}{(\partial y)^2} f_x - i\sqrt{2}v^2 f_y^2 \frac{\partial}{\partial y} f_x \frac{\partial^2}{(\partial y)^2} f_x \\
& + \sqrt{2}v^2 f_x^2 \frac{\partial}{\partial x} f_y \frac{\partial^2}{(\partial x)^2} f_y + \sqrt{2}v^2 f_y^2 \frac{\partial}{\partial x} f_y \frac{\partial^2}{(\partial x)^2} f_y - (i-1)\sqrt{2}v^2 f_x^2 \frac{\partial}{\partial x} f_y \frac{\partial^2}{\partial x \partial y} f_y - (i-1)\sqrt{2}v^2 f_y^2 \frac{\partial}{\partial x} f_y \frac{\partial^2}{\partial x \partial y} f_y + \sqrt{2}v^2 f_x^2 \frac{\partial^2}{(\partial x)^2} f_y \frac{\partial}{\partial y} f_y + \sqrt{2}v^2 f_y^2 \frac{\partial^2}{(\partial x)^2} f_y \frac{\partial}{\partial y} f_y \\
& - (i-1)\sqrt{2}v^2 f_x^2 \frac{\partial^2}{\partial x \partial y} f_y \frac{\partial}{\partial y} f_y - (i-1)\sqrt{2}v^2 f_y^2 \frac{\partial^2}{\partial x \partial y} f_y \frac{\partial}{\partial y} f_y - i\sqrt{2}v^2 f_x^2 \frac{\partial}{\partial x} f_y \frac{\partial^2}{(\partial y)^2} f_y - i\sqrt{2}v^2 f_y^2 \frac{\partial}{\partial x} f_y \frac{\partial^2}{(\partial y)^2} f_y - i\sqrt{2}v^2 f_x^2 \frac{\partial}{\partial y} f_y \frac{\partial^2}{(\partial y)^2} f_y - i\sqrt{2}v^2 f_y^2 \frac{\partial}{\partial y} f_y \frac{\partial^2}{(\partial y)^2} f_y \\
& + \sqrt{2}v^2 \frac{\partial}{\partial x} f_x \frac{\partial^2}{(\partial x)^2} f_x - (i-1)\sqrt{2}v^2 \frac{\partial}{\partial x} f_x \frac{\partial^2}{\partial x \partial y} f_x + \sqrt{2}v^2 \frac{\partial^2}{(\partial x)^2} f_x \frac{\partial}{\partial y} f_x - (i-1)\sqrt{2}v^2 \frac{\partial^2}{\partial x \partial y} f_x \frac{\partial}{\partial y} f_x - i\sqrt{2}v^2 \frac{\partial}{\partial x} f_x \frac{\partial^2}{(\partial y)^2} f_x - i\sqrt{2}v^2 \frac{\partial}{\partial y} f_x \frac{\partial^2}{(\partial y)^2} f_x \\
& + \sqrt{2}v^2 \frac{\partial}{\partial x} f_y \frac{\partial^2}{(\partial x)^2} f_y - (i-1)\sqrt{2}v^2 \frac{\partial}{\partial x} f_y \frac{\partial^2}{\partial x \partial y} f_y + \sqrt{2}v^2 \frac{\partial^2}{(\partial x)^2} f_y \frac{\partial}{\partial y} f_y - (i-1)\sqrt{2}v^2 \frac{\partial^2}{\partial x \partial y} f_y \frac{\partial}{\partial y} f_y - i\sqrt{2}v^2 \frac{\partial}{\partial x} f_y \frac{\partial^2}{(\partial y)^2} f_y - i\sqrt{2}v^2 \frac{\partial}{\partial y} f_y \frac{\partial^2}{(\partial y)^2} f_y
\end{aligned}$$

$$\alpha = \frac{1}{2}\sqrt{2} f_x \frac{\partial}{\partial x} f_x - \frac{1}{2} i \sqrt{2} f_x \frac{\partial}{\partial y} f_x + \frac{1}{2}\sqrt{2} f_y \frac{\partial}{\partial x} f_y - \frac{1}{2} i \sqrt{2} f_y \frac{\partial}{\partial y} f_y$$

$$\beta = -\frac{1}{2}\sqrt{2} f_x \frac{\partial}{\partial x} f_x - \frac{1}{2} i \sqrt{2} f_x \frac{\partial}{\partial y} f_x - \frac{1}{2}\sqrt{2} f_y \frac{\partial}{\partial x} f_y - \frac{1}{2} i \sqrt{2} f_y \frac{\partial}{\partial y} f_y$$

$$\tau = 0$$

$$\pi = 0$$

$\epsilon = 0$

$$\gamma = -v\frac{\partial}{\partial x}f_x^2 - 2\,v\frac{\partial}{\partial x}f_x\frac{\partial}{\partial y}f_x - v\frac{\partial}{\partial y}f_x^2 - v\frac{\partial}{\partial x}f_y^2 - 2\,v\frac{\partial}{\partial x}f_y\frac{\partial}{\partial y}f_y - v\frac{\partial}{\partial y}f_y^2 - \frac{1}{2}\,\frac{\partial}{\partial v}H^0(u,v)$$

Ricci Tensor Components

$\Phi_{00} = 0$

$\Phi_{01} = 0$

$\Phi_{02} = 0$

$\Phi_{10} = 0$

$$\begin{aligned}\Phi_{11} = &-\frac{1}{2}\,f_x^2\frac{\partial}{\partial x}f_x^2 + \frac{1}{2}\,f_y^2\frac{\partial}{\partial x}f_x^2 + \frac{1}{2}\,f_x^3\frac{\partial^2}{(\partial x)^2}f_x + \frac{1}{2}\,f_xf_y^2\frac{\partial^2}{(\partial x)^2}f_x - \frac{1}{2}\,f_x^2\frac{\partial}{\partial y}f_x^2 + \frac{1}{2}\,f_y^2\frac{\partial}{\partial y}f_x^2 + \frac{1}{2}\,f_x^3\frac{\partial^2}{(\partial y)^2}f_x + \frac{1}{2}\,f_xf_y^2\frac{\partial^2}{(\partial y)^2}f_x - 2\,f_xf_y\frac{\partial}{\partial x}f_x\frac{\partial}{\partial x}f_y\\
&+\frac{1}{2}\,f_x^2\frac{\partial}{\partial x}f_y^2 - \frac{1}{2}\,f_y^2\frac{\partial}{\partial x}f_y^2 + \frac{1}{2}\,f_x^2f_y\frac{\partial^2}{(\partial x)^2}f_y + \frac{1}{2}\,f_y^3\frac{\partial^2}{(\partial x)^2}f_y - 2\,f_xf_y\frac{\partial}{\partial y}f_x\frac{\partial}{\partial y}f_y + \frac{1}{2}\,f_x^2\frac{\partial}{\partial y}f_y^2 - \frac{1}{2}\,f_y^2\frac{\partial}{\partial y}f_y^2 + \frac{1}{2}\,f_x^2f_y\frac{\partial^2}{(\partial y)^2}f_y + \frac{1}{2}\,f_y^3\frac{\partial^2}{(\partial y)^2}f_y\\
&+\frac{\partial}{\partial x}f_x^2 + \frac{1}{2}\,f_x\frac{\partial^2}{(\partial x)^2}f_x + \frac{\partial}{\partial x}f_x\frac{\partial}{\partial y}f_x + \frac{\partial}{\partial y}f_x^2 + \frac{1}{2}\,f_x\frac{\partial^2}{(\partial y)^2}f_x + \frac{\partial}{\partial x}f_y^2 + \frac{1}{2}\,f_y\frac{\partial^2}{(\partial x)^2}f_y + \frac{\partial}{\partial x}f_y\frac{\partial}{\partial y}f_y + \frac{\partial}{\partial y}f_y^2 + \frac{1}{2}\,f_y\frac{\partial^2}{(\partial y)^2}f_y + \frac{1}{4}\,\frac{\partial^2}{(\partial v)^2}H^0(u,v)\end{aligned}$$

$$\begin{aligned}\Phi_{12} = &-\sqrt{2}vf_x^2\frac{\partial}{\partial x}f_x\frac{\partial^2}{(\partial x)^2}f_x - \sqrt{2}vf_y^2\frac{\partial}{\partial x}f_x\frac{\partial^2}{(\partial x)^2}f_x - (i+1)\,\sqrt{2}vf_x^2\frac{\partial}{\partial x}f_x\frac{\partial^2}{\partial x\partial y}f_x - (i+1)\,\sqrt{2}vf_y^2\frac{\partial}{\partial x}f_x\frac{\partial^2}{\partial x\partial y}f_x - \sqrt{2}vf_x^2\frac{\partial^2}{(\partial x)^2}f_x\frac{\partial}{\partial y}f_x - \sqrt{2}vf_y^2\frac{\partial^2}{(\partial x)^2}f_x\frac{\partial}{\partial y}f_x\\
&-(i+1)\,\sqrt{2}vf_x^2\frac{\partial^2}{\partial x\partial y}f_x\frac{\partial}{\partial y}f_x - (i+1)\,\sqrt{2}vf_y^2\frac{\partial^2}{\partial x\partial y}f_x\frac{\partial}{\partial y}f_x - i\,\sqrt{2}vf_x^2\frac{\partial}{\partial x}f_x\frac{\partial^2}{(\partial y)^2}f_x - i\,\sqrt{2}vf_y^2\frac{\partial}{\partial x}f_x\frac{\partial^2}{(\partial y)^2}f_x - i\,\sqrt{2}vf_x^2\frac{\partial}{\partial y}f_x\frac{\partial^2}{(\partial y)^2}f_x - i\,\sqrt{2}vf_y^2\frac{\partial}{\partial y}f_x\frac{\partial^2}{(\partial y)^2}f_x\\
&-\sqrt{2}vf_x^2\frac{\partial}{\partial x}f_y\frac{\partial^2}{(\partial x)^2}f_y - \sqrt{2}vf_y^2\frac{\partial}{\partial x}f_y\frac{\partial^2}{(\partial x)^2}f_y - (i+1)\,\sqrt{2}vf_x^2\frac{\partial}{\partial x}f_y\frac{\partial^2}{\partial x\partial y}f_y - (i+1)\,\sqrt{2}vf_y^2\frac{\partial}{\partial x}f_y\frac{\partial^2}{\partial x\partial y}f_y - \sqrt{2}vf_x^2\frac{\partial^2}{(\partial x)^2}f_y\frac{\partial}{\partial y}f_y - \sqrt{2}vf_y^2\frac{\partial^2}{(\partial x)^2}f_y\frac{\partial}{\partial y}f_y\\
&-(i+1)\,\sqrt{2}vf_x^2\frac{\partial^2}{\partial x\partial y}f_y\frac{\partial}{\partial y}f_y - (i+1)\,\sqrt{2}vf_y^2\frac{\partial^2}{\partial x\partial y}f_y\frac{\partial}{\partial y}f_y - i\,\sqrt{2}vf_x^2\frac{\partial}{\partial x}f_y\frac{\partial^2}{(\partial y)^2}f_y - i\,\sqrt{2}vf_y^2\frac{\partial}{\partial x}f_y\frac{\partial^2}{(\partial y)^2}f_y - i\,\sqrt{2}vf_x^2\frac{\partial}{\partial y}f_y\frac{\partial^2}{(\partial y)^2}f_y - i\,\sqrt{2}vf_y^2\frac{\partial}{\partial y}f_y\frac{\partial^2}{(\partial y)^2}f_y\\
&-\sqrt{2}v\frac{\partial}{\partial x}f_x\frac{\partial^2}{(\partial x)^2}f_x - (i+1)\,\sqrt{2}v\frac{\partial}{\partial x}f_x\frac{\partial^2}{\partial x\partial y}f_x - \sqrt{2}v\frac{\partial^2}{(\partial x)^2}f_x\frac{\partial}{\partial y}f_x - (i+1)\,\sqrt{2}v\frac{\partial^2}{\partial x\partial y}f_x\frac{\partial}{\partial y}f_x - i\,\sqrt{2}v\frac{\partial}{\partial x}f_x\frac{\partial^2}{(\partial y)^2}f_x - i\,\sqrt{2}v\frac{\partial}{\partial y}f_x\frac{\partial^2}{(\partial y)^2}f_x\\
&-\sqrt{2}v\frac{\partial}{\partial x}f_y\frac{\partial^2}{(\partial x)^2}f_y - (i+1)\,\sqrt{2}v\frac{\partial}{\partial x}f_y\frac{\partial^2}{\partial x\partial y}f_y - \sqrt{2}v\frac{\partial^2}{(\partial x)^2}f_y\frac{\partial}{\partial y}f_y - (i+1)\,\sqrt{2}v\frac{\partial^2}{\partial x\partial y}f_y\frac{\partial}{\partial y}f_y - i\,\sqrt{2}v\frac{\partial}{\partial x}f_y\frac{\partial^2}{(\partial y)^2}f_y - i\,\sqrt{2}v\frac{\partial}{\partial y}f_y\frac{\partial^2}{(\partial y)^2}f_y\end{aligned}$$

$$\Phi_{20} = 0$$

$$\begin{aligned}
\Phi_{21} = & -\sqrt{2}vf_x^2\frac{\partial}{\partial x}f_x\frac{\partial^2}{(\partial x)^2}f_x - \sqrt{2}vf_y^2\frac{\partial}{\partial x}f_x\frac{\partial^2}{(\partial x)^2}f_x + (i-1)\sqrt{2}vf_x^2\frac{\partial}{\partial x}f_x\frac{\partial^2}{\partial x\partial y}f_x + (i-1)\sqrt{2}vf_y^2\frac{\partial}{\partial x}f_x\frac{\partial^2}{\partial x\partial y}f_x - \sqrt{2}vf_x^2\frac{\partial^2}{(\partial x)^2}f_x\frac{\partial}{\partial y}f_x - \sqrt{2}vf_y^2\frac{\partial^2}{(\partial x)^2}f_x\frac{\partial}{\partial y}f_x \\
& + (i-1)\sqrt{2}vf_x^2\frac{\partial^2}{\partial x\partial y}f_x\frac{\partial}{\partial y}f_x + (i-1)\sqrt{2}vf_y^2\frac{\partial^2}{\partial x\partial y}f_x\frac{\partial}{\partial y}f_x + i\sqrt{2}vf_x^2\frac{\partial}{\partial x}f_x\frac{\partial^2}{(\partial y)^2}f_x + i\sqrt{2}vf_y^2\frac{\partial}{\partial x}f_x\frac{\partial^2}{(\partial y)^2}f_x + i\sqrt{2}vf_x^2\frac{\partial}{\partial y}f_x\frac{\partial^2}{(\partial y)^2}f_x + i\sqrt{2}vf_y^2\frac{\partial}{\partial y}f_x\frac{\partial^2}{(\partial y)^2}f_x \\
& - \sqrt{2}vf_x^2\frac{\partial}{\partial x}f_y\frac{\partial^2}{(\partial x)^2}f_y - \sqrt{2}vf_y^2\frac{\partial}{\partial x}f_y\frac{\partial^2}{(\partial x)^2}f_y + (i-1)\sqrt{2}vf_x^2\frac{\partial}{\partial x}f_y\frac{\partial^2}{\partial x\partial y}f_y + (i-1)\sqrt{2}vf_y^2\frac{\partial}{\partial x}f_y\frac{\partial^2}{\partial x\partial y}f_y - \sqrt{2}vf_x^2\frac{\partial^2}{(\partial x)^2}f_y\frac{\partial}{\partial y}f_y - \sqrt{2}vf_y^2\frac{\partial^2}{(\partial x)^2}f_y\frac{\partial}{\partial y}f_y \\
& + (i-1)\sqrt{2}vf_x^2\frac{\partial^2}{\partial x\partial y}f_y\frac{\partial}{\partial y}f_y + (i-1)\sqrt{2}vf_y^2\frac{\partial^2}{\partial x\partial y}f_y\frac{\partial}{\partial y}f_y + i\sqrt{2}vf_x^2\frac{\partial}{\partial x}f_y\frac{\partial^2}{(\partial y)^2}f_y + i\sqrt{2}vf_y^2\frac{\partial}{\partial x}f_y\frac{\partial^2}{(\partial y)^2}f_y + i\sqrt{2}vf_x^2\frac{\partial}{\partial y}f_y\frac{\partial^2}{(\partial y)^2}f_y + i\sqrt{2}vf_y^2\frac{\partial}{\partial y}f_y\frac{\partial^2}{(\partial y)^2}f_y \\
& - \sqrt{2}v\frac{\partial}{\partial x}f_x\frac{\partial^2}{(\partial x)^2}f_x + (i-1)\sqrt{2}v\frac{\partial}{\partial x}f_x\frac{\partial^2}{\partial x\partial y}f_x - \sqrt{2}v\frac{\partial^2}{(\partial x)^2}f_x\frac{\partial}{\partial y}f_x + (i-1)\sqrt{2}v\frac{\partial^2}{\partial x\partial y}f_x\frac{\partial}{\partial y}f_x + i\sqrt{2}v\frac{\partial}{\partial x}f_x\frac{\partial^2}{(\partial y)^2}f_x + i\sqrt{2}v\frac{\partial}{\partial y}f_x\frac{\partial^2}{(\partial y)^2}f_x \\
& - \sqrt{2}v\frac{\partial}{\partial x}f_y\frac{\partial^2}{(\partial x)^2}f_y + (i-1)\sqrt{2}v\frac{\partial}{\partial x}f_y\frac{\partial^2}{\partial x\partial y}f_y - \sqrt{2}v\frac{\partial^2}{(\partial x)^2}f_y\frac{\partial}{\partial y}f_y + (i-1)\sqrt{2}v\frac{\partial^2}{\partial x\partial y}f_y\frac{\partial}{\partial y}f_y + i\sqrt{2}v\frac{\partial}{\partial x}f_y\frac{\partial^2}{(\partial y)^2}f_y + i\sqrt{2}v\frac{\partial}{\partial y}f_y\frac{\partial^2}{(\partial y)^2}f_y
\end{aligned}$$

$$
\begin{aligned}
\Phi_{22} = {}& v^2 f_x^4 \frac{\partial^2}{(\partial x)^2} f_x^2 + 2\, v^2 f_x^2 f_y^2 \frac{\partial^2}{(\partial x)^2} f_x^2 + v^2 f_y^4 \frac{\partial^2}{(\partial x)^2} f_x^2 + v^2 f_x^4 \frac{\partial}{\partial x} f_x \frac{\partial^3}{(\partial x)^3} f_x + 2\, v^2 f_x^2 f_y^2 \frac{\partial}{\partial x} f_x \frac{\partial^3}{(\partial x)^3} f_x + v^2 f_y^4 \frac{\partial}{\partial x} f_x \frac{\partial^3}{(\partial x)^3} f_x + v^2 f_x^4 \frac{\partial}{\partial x} f_x \frac{\partial^3}{(\partial x)^2 \partial y} f_x + 2\, v^2 f_x^2 f_y^2 \frac{\partial}{\partial x} f_x \frac{\partial^3}{(\partial x)^2 \partial y} f_x + v^2 f_y^4 \frac{\partial}{\partial x} f_x \frac{\partial^3}{(\partial x)^2 \partial y} f_x \\
& + 2\, v^2 f_x^4 \frac{\partial^2}{(\partial x)^2} f_x \frac{\partial^2}{\partial x \partial y} f_x + 4\, v^2 f_x^2 f_y^2 \frac{\partial^2}{(\partial x)^2} f_x \frac{\partial^2}{\partial x \partial y} f_x + 2\, v^2 f_y^4 \frac{\partial^2}{(\partial x)^2} f_x \frac{\partial^2}{\partial x \partial y} f_x + 2\, v^2 f_x^4 \frac{\partial^2}{\partial x \partial y} f_x^2 + 4\, v^2 f_x^2 f_y^2 \frac{\partial^2}{\partial x \partial y} f_x^2 + 2\, v^2 f_y^4 \frac{\partial^2}{\partial x \partial y} f_x^2 + v^2 f_x^4 \frac{\partial}{\partial x} f_x \frac{\partial^3}{\partial x (\partial y)^2} f_x + 2\, v^2 f_x^2 f_y^2 \frac{\partial}{\partial x} f_x \frac{\partial^3}{\partial x (\partial y)^2} f_x \\
& + v^2 f_y^4 \frac{\partial}{\partial x} f_x \frac{\partial^3}{\partial x (\partial y)^2} f_x + v^2 f_x^4 \frac{\partial^3}{(\partial x)^3} f_x \frac{\partial}{\partial y} f_x + 2\, v^2 f_x^2 f_y^2 \frac{\partial^3}{(\partial x)^3} f_x \frac{\partial}{\partial y} f_x + v^2 f_y^4 \frac{\partial^3}{(\partial x)^3} f_x \frac{\partial}{\partial y} f_x + v^2 f_x^4 \frac{\partial^3}{(\partial x)^2 \partial y} f_x \frac{\partial}{\partial y} f_x + 2\, v^2 f_x^2 f_y^2 \frac{\partial^3}{(\partial x)^2 \partial y} f_x \frac{\partial}{\partial y} f_x + v^2 f_y^4 \frac{\partial^3}{(\partial x)^2 \partial y} f_x \frac{\partial}{\partial y} f_x + v^2 f_x^4 \frac{\partial^3}{\partial x (\partial y)^2} f_x \frac{\partial}{\partial y} f_x \\
& + 2\, v^2 f_x^2 f_y^2 \frac{\partial^3}{\partial x (\partial y)^2} f_x \frac{\partial}{\partial y} f_x + v^2 f_y^4 \frac{\partial^3}{\partial x (\partial y)^2} f_x \frac{\partial}{\partial y} f_x + 2\, v^2 f_x^4 \frac{\partial^2}{\partial x \partial y} f_x \frac{\partial^2}{(\partial y)^2} f_x + 4\, v^2 f_x^2 f_y^2 \frac{\partial^2}{\partial x \partial y} f_x \frac{\partial^2}{(\partial y)^2} f_x + 2\, v^2 f_y^4 \frac{\partial^2}{\partial x \partial y} f_x \frac{\partial^2}{(\partial y)^2} f_x + v^2 f_x^4 \frac{\partial^2}{(\partial y)^2} f_x^2 + 2\, v^2 f_x^2 f_y^2 \frac{\partial^2}{(\partial y)^2} f_x^2 + v^2 f_y^4 \frac{\partial^2}{(\partial y)^2} f_x^2 \\
& + v^2 f_x^4 \frac{\partial}{\partial x} f_x \frac{\partial^3}{(\partial y)^3} f_x + 2\, v^2 f_x^2 f_y^2 \frac{\partial}{\partial x} f_x \frac{\partial^3}{(\partial y)^3} f_x + v^2 f_y^4 \frac{\partial}{\partial x} f_x \frac{\partial^3}{(\partial y)^3} f_x + v^2 f_x^4 \frac{\partial}{\partial y} f_x \frac{\partial^3}{(\partial y)^3} f_x + 2\, v^2 f_x^2 f_y^2 \frac{\partial}{\partial y} f_x \frac{\partial^3}{(\partial y)^3} f_x + v^2 f_y^4 \frac{\partial}{\partial y} f_x \frac{\partial^3}{(\partial y)^3} f_x + v^2 f_x^4 \frac{\partial^2}{(\partial x)^2} f_y^2 + 2\, v^2 f_x^2 f_y^2 \frac{\partial^2}{(\partial x)^2} f_y^2 + v^2 f_y^4 \frac{\partial^2}{(\partial x)^2} f_y^2 \\
& + v^2 f_x^4 \frac{\partial}{\partial x} f_y \frac{\partial^3}{(\partial x)^3} f_y + 2\, v^2 f_x^2 f_y^2 \frac{\partial}{\partial x} f_y \frac{\partial^3}{(\partial x)^3} f_y + v^2 f_y^4 \frac{\partial}{\partial x} f_y \frac{\partial^3}{(\partial x)^3} f_y + v^2 f_x^4 \frac{\partial}{\partial x} f_y \frac{\partial^3}{(\partial x)^2 \partial y} f_y + 2\, v^2 f_x^2 f_y^2 \frac{\partial}{\partial x} f_y \frac{\partial^3}{(\partial x)^2 \partial y} f_y + v^2 f_y^4 \frac{\partial}{\partial x} f_y \frac{\partial^3}{(\partial x)^2 \partial y} f_y + 2\, v^2 f_x^4 \frac{\partial^2}{(\partial x)^2} f_y \frac{\partial^2}{\partial x \partial y} f_y + 4\, v^2 f_x^2 f_y^2 \frac{\partial^2}{(\partial x)^2} f_y \frac{\partial^2}{\partial x \partial y} f_y \\
& + 2\, v^2 f_y^4 \frac{\partial^2}{(\partial x)^2} f_y \frac{\partial^2}{\partial x \partial y} f_y + 2\, v^2 f_x^4 \frac{\partial^2}{\partial x \partial y} f_y^2 + 4\, v^2 f_x^2 f_y^2 \frac{\partial^2}{\partial x \partial y} f_y^2 + 2\, v^2 f_y^4 \frac{\partial^2}{\partial x \partial y} f_y^2 + v^2 f_x^4 \frac{\partial}{\partial x} f_y \frac{\partial^3}{\partial x (\partial y)^2} f_y + 2\, v^2 f_x^2 f_y^2 \frac{\partial}{\partial x} f_y \frac{\partial^3}{\partial x (\partial y)^2} f_y + v^2 f_y^4 \frac{\partial}{\partial x} f_y \frac{\partial^3}{\partial x (\partial y)^2} f_y + v^2 f_x^4 \frac{\partial^3}{(\partial x)^3} f_y \frac{\partial}{\partial y} f_y + 2\, v^2 f_x^2 f_y^2 \frac{\partial^3}{(\partial x)^3} f_y \frac{\partial}{\partial y} f_y \\
& + v^2 f_y^4 \frac{\partial^3}{(\partial x)^3} f_y \frac{\partial}{\partial y} f_y + v^2 f_x^4 \frac{\partial^3}{(\partial x)^2 \partial y} f_y \frac{\partial}{\partial y} f_y + 2\, v^2 f_x^2 f_y^2 \frac{\partial^3}{(\partial x)^2 \partial y} f_y \frac{\partial}{\partial y} f_y + v^2 f_y^4 \frac{\partial^3}{(\partial x)^2 \partial y} f_y \frac{\partial}{\partial y} f_y + v^2 f_x^4 \frac{\partial^3}{\partial x (\partial y)^2} f_y \frac{\partial}{\partial y} f_y + 2\, v^2 f_x^2 f_y^2 \frac{\partial^3}{\partial x (\partial y)^2} f_y \frac{\partial}{\partial y} f_y + v^2 f_y^4 \frac{\partial^3}{\partial x (\partial y)^2} f_y \frac{\partial}{\partial y} f_y + 2\, v^2 f_x^4 \frac{\partial^2}{\partial x \partial y} f_y \frac{\partial^2}{(\partial y)^2} f_y \\
& + 4\, v^2 f_x^2 f_y^2 \frac{\partial^2}{\partial x \partial y} f_y \frac{\partial^2}{(\partial y)^2} f_y + 2\, v^2 f_y^4 \frac{\partial^2}{\partial x \partial y} f_y \frac{\partial^2}{(\partial y)^2} f_y + v^2 f_x^4 \frac{\partial^2}{(\partial y)^2} f_y^2 + 2\, v^2 f_x^2 f_y^2 \frac{\partial^2}{(\partial y)^2} f_y^2 + v^2 f_y^4 \frac{\partial^2}{(\partial y)^2} f_y^2 + v^2 f_x^4 \frac{\partial}{\partial x} f_y \frac{\partial^3}{(\partial y)^3} f_y + 2\, v^2 f_x^2 f_y^2 \frac{\partial}{\partial x} f_y \frac{\partial^3}{(\partial y)^3} f_y + v^2 f_y^4 \frac{\partial}{\partial x} f_y \frac{\partial^3}{(\partial y)^3} f_y + v^2 f_x^4 \frac{\partial}{\partial y} f_y \frac{\partial^3}{(\partial y)^3} f_y \\
& + 2\, v^2 f_x^2 f_y^2 \frac{\partial}{\partial y} f_y \frac{\partial^3}{(\partial y)^3} f_y + v^2 f_y^4 \frac{\partial}{\partial y} f_y \frac{\partial^3}{(\partial y)^3} f_y + 2\, v^2 f_x^2 \frac{\partial^2}{(\partial x)^2} f_x^2 + 2\, v^2 f_y^2 \frac{\partial^2}{(\partial x)^2} f_x^2 + 2\, v^2 f_x^2 \frac{\partial}{\partial x} f_x \frac{\partial^3}{(\partial x)^3} f_x + 2\, v^2 f_y^2 \frac{\partial}{\partial x} f_x \frac{\partial^3}{(\partial x)^3} f_x + 2\, v^2 f_x^2 \frac{\partial}{\partial x} f_x \frac{\partial^3}{(\partial x)^2 \partial y} f_x + 2\, v^2 f_y^2 \frac{\partial}{\partial x} f_x \frac{\partial^3}{(\partial x)^2 \partial y} f_x + 4\, v^2 f_x^2 \frac{\partial^2}{(\partial x)^2} f_x \frac{\partial^2}{\partial x \partial y} f_x \\
& + 4\, v^2 f_y^2 \frac{\partial^2}{(\partial x)^2} f_x \frac{\partial^2}{\partial x \partial y} f_x + 4\, v^2 f_x^2 \frac{\partial^2}{\partial x \partial y} f_x^2 + 4\, v^2 f_y^2 \frac{\partial^2}{\partial x \partial y} f_x^2 + 2\, v^2 f_x^2 \frac{\partial}{\partial x} f_x \frac{\partial^3}{\partial x (\partial y)^2} f_x + 2\, v^2 f_y^2 \frac{\partial}{\partial x} f_x \frac{\partial^3}{\partial x (\partial y)^2} f_x + 2\, v^2 f_x^2 \frac{\partial^3}{(\partial x)^3} f_x \frac{\partial}{\partial y} f_x + 2\, v^2 f_y^2 \frac{\partial^3}{(\partial x)^3} f_x \frac{\partial}{\partial y} f_x + 2\, v^2 f_x^2 \frac{\partial^3}{(\partial x)^2 \partial y} f_x \frac{\partial}{\partial y} f_x \\
& + 2\, v^2 f_y^2 \frac{\partial^3}{(\partial x)^2 \partial y} f_x \frac{\partial}{\partial y} f_x + 2\, v^2 f_x^2 \frac{\partial^3}{\partial x (\partial y)^2} f_x \frac{\partial}{\partial y} f_x + 2\, v^2 f_y^2 \frac{\partial^3}{\partial x (\partial y)^2} f_x \frac{\partial}{\partial y} f_x + 4\, v^2 f_x^2 \frac{\partial^2}{\partial x \partial y} f_x \frac{\partial^2}{(\partial y)^2} f_x + 4\, v^2 f_y^2 \frac{\partial^2}{\partial x \partial y} f_x \frac{\partial^2}{(\partial y)^2} f_x + 2\, v^2 f_x^2 \frac{\partial^2}{(\partial y)^2} f_x^2 + 2\, v^2 f_y^2 \frac{\partial^2}{(\partial y)^2} f_x^2 + 2\, v^2 f_x^2 \frac{\partial}{\partial x} f_x \frac{\partial^3}{(\partial y)^3} f_x \\
& + 2\, v^2 f_y^2 \frac{\partial}{\partial x} f_x \frac{\partial^3}{(\partial y)^3} f_x + 2\, v^2 f_x^2 \frac{\partial}{\partial y} f_x \frac{\partial^3}{(\partial y)^3} f_x + 2\, v^2 f_y^2 \frac{\partial}{\partial y} f_x \frac{\partial^3}{(\partial y)^3} f_x + 2\, v^2 f_x^2 \frac{\partial^2}{(\partial x)^2} f_y^2 + 2\, v^2 f_y^2 \frac{\partial^2}{(\partial x)^2} f_y^2 + 2\, v^2 f_x^2 \frac{\partial}{\partial x} f_y \frac{\partial^3}{(\partial x)^3} f_y + 2\, v^2 f_y^2 \frac{\partial}{\partial x} f_y \frac{\partial^3}{(\partial x)^3} f_y + 2\, v^2 f_x^2 \frac{\partial}{\partial x} f_y \frac{\partial^3}{(\partial x)^2 \partial y} f_y + 2\, v^2 f_y^2 \frac{\partial}{\partial x} f_y \frac{\partial^3}{(\partial x)^2 \partial y} f_y \\
& + 4\, v^2 f_x^2 \frac{\partial^2}{(\partial x)^2} f_y \frac{\partial^2}{\partial x \partial y} f_y + 4\, v^2 f_y^2 \frac{\partial^2}{(\partial x)^2} f_y \frac{\partial^2}{\partial x \partial y} f_y + 4\, v^2 f_x^2 \frac{\partial^2}{\partial x \partial y} f_y^2 + 4\, v^2 f_y^2 \frac{\partial^2}{\partial x \partial y} f_y^2 + 2\, v^2 f_x^2 \frac{\partial}{\partial x} f_y \frac{\partial^3}{\partial x (\partial y)^2} f_y + 2\, v^2 f_y^2 \frac{\partial}{\partial x} f_y \frac{\partial^3}{\partial x (\partial y)^2} f_y + 2\, v^2 f_x^2 \frac{\partial^3}{(\partial x)^3} f_y \frac{\partial}{\partial y} f_y + 2\, v^2 f_y^2 \frac{\partial^3}{(\partial x)^3} f_y \frac{\partial}{\partial y} f_y \\
& + 2\, v^2 f_x^2 \frac{\partial^3}{(\partial x)^2 \partial y} f_y \frac{\partial}{\partial y} f_y + 2\, v^2 f_y^2 \frac{\partial^3}{(\partial x)^2 \partial y} f_y \frac{\partial}{\partial y} f_y + 2\, v^2 f_x^2 \frac{\partial^3}{\partial x (\partial y)^2} f_y \frac{\partial}{\partial y} f_y + 2\, v^2 f_y^2 \frac{\partial^3}{\partial x (\partial y)^2} f_y \frac{\partial}{\partial y} f_y + 4\, v^2 f_x^2 \frac{\partial^2}{\partial x \partial y} f_y \frac{\partial^2}{(\partial y)^2} f_y + 4\, v^2 f_y^2 \frac{\partial^2}{\partial x \partial y} f_y \frac{\partial^2}{(\partial y)^2} f_y + 2\, v^2 f_x^2 \frac{\partial^2}{(\partial y)^2} f_y^2 + 2\, v^2 f_y^2 \frac{\partial^2}{(\partial y)^2} f_y^2 \\
& + 2\, v^2 f_x^2 \frac{\partial}{\partial x} f_y \frac{\partial^3}{(\partial y)^3} f_y + 2\, v^2 f_y^2 \frac{\partial}{\partial x} f_y \frac{\partial^3}{(\partial y)^3} f_y + 2\, v^2 f_x^2 \frac{\partial}{\partial y} f_y \frac{\partial^3}{(\partial y)^3} f_y + 2\, v^2 f_y^2 \frac{\partial}{\partial y} f_y \frac{\partial^3}{(\partial y)^3} f_y + v^2 \frac{\partial^2}{(\partial x)^2} f_x^2 + v^2 \frac{\partial}{\partial x} f_x \frac{\partial^3}{(\partial x)^3} f_x + v^2 \frac{\partial}{\partial x} f_x \frac{\partial^3}{(\partial x)^2 \partial y} f_x + 2\, v^2 \frac{\partial^2}{(\partial x)^2} f_x \frac{\partial^2}{\partial x \partial y} f_x + 2\, v^2 \frac{\partial^2}{\partial x \partial y} f_x^2 + v^2 \frac{\partial}{\partial x} f_x \frac{\partial^3}{\partial x (\partial y)^2} f_x \\
& + v^2 \frac{\partial^3}{(\partial x)^3} f_x \frac{\partial}{\partial y} f_x + v^2 \frac{\partial^3}{(\partial x)^2 \partial y} f_x \frac{\partial}{\partial y} f_x + v^2 \frac{\partial^3}{\partial x (\partial y)^2} f_x \frac{\partial}{\partial y} f_x + 2\, v^2 \frac{\partial^2}{\partial x \partial y} f_x \frac{\partial^2}{(\partial y)^2} f_x + v^2 \frac{\partial^2}{(\partial y)^2} f_x^2 + v^2 \frac{\partial}{\partial x} f_x \frac{\partial^3}{(\partial y)^3} f_x + v^2 \frac{\partial}{\partial y} f_x \frac{\partial^3}{(\partial y)^3} f_x + v^2 \frac{\partial^2}{(\partial x)^2} f_y^2 + v^2 \frac{\partial}{\partial x} f_y \frac{\partial^3}{(\partial x)^3} f_y + v^2 \frac{\partial}{\partial x} f_y \frac{\partial^3}{(\partial x)^2 \partial y} f_y \\
& + 2\, v^2 \frac{\partial^2}{(\partial x)^2} f_y \frac{\partial^2}{\partial x \partial y} f_y + 2\, v^2 \frac{\partial^2}{\partial x \partial y} f_y^2 + v^2 \frac{\partial}{\partial x} f_y \frac{\partial^3}{\partial x (\partial y)^2} f_y + v^2 \frac{\partial^3}{(\partial x)^3} f_y \frac{\partial}{\partial y} f_y + v^2 \frac{\partial^3}{(\partial x)^2 \partial y} f_y \frac{\partial}{\partial y} f_y + v^2 \frac{\partial^3}{\partial x (\partial y)^2} f_y \frac{\partial}{\partial y} f_y + 2\, v^2 \frac{\partial^2}{\partial x \partial y} f_y \frac{\partial^2}{(\partial y)^2} f_y + v^2 \frac{\partial^2}{(\partial y)^2} f_y^2 + v^2 \frac{\partial}{\partial x} f_y \frac{\partial^3}{(\partial y)^3} f_y + v^2 \frac{\partial}{\partial y} f_y \frac{\partial^3}{(\partial y)^3} f_y
\end{aligned}
$$

$$
\begin{aligned}
\Lambda = {}& -\frac{1}{6} f_x^2 \frac{\partial}{\partial x} f_x^2 + \frac{1}{6} f_y^2 \frac{\partial}{\partial x} f_x^2 + \frac{1}{6} f_x^3 \frac{\partial^2}{(\partial x)^2} f_x + \frac{1}{6} f_x f_y^2 \frac{\partial^2}{(\partial x)^2} f_x - \frac{1}{6} f_x^2 \frac{\partial}{\partial y} f_x^2 + \frac{1}{6} f_y^2 \frac{\partial}{\partial y} f_x^2 + \frac{1}{6} f_x^3 \frac{\partial^2}{(\partial y)^2} f_x + \frac{1}{6} f_x f_y^2 \frac{\partial^2}{(\partial y)^2} f_x - \frac{2}{3} f_x f_y \frac{\partial}{\partial x} f_x \frac{\partial}{\partial x} f_y \\
& + \frac{1}{6} f_x^2 \frac{\partial}{\partial x} f_y^2 - \frac{1}{6} f_y^2 \frac{\partial}{\partial x} f_y^2 + \frac{1}{6} f_x^2 f_y \frac{\partial^2}{(\partial x)^2} f_y + \frac{1}{6} f_y^3 \frac{\partial^2}{(\partial x)^2} f_y - \frac{2}{3} f_x f_y \frac{\partial}{\partial y} f_x \frac{\partial}{\partial y} f_y + \frac{1}{6} f_x^2 \frac{\partial}{\partial y} f_y^2 - \frac{1}{6} f_y^2 \frac{\partial}{\partial y} f_y^2 + \frac{1}{6} f_x^2 f_y \frac{\partial^2}{(\partial y)^2} f_y \\
& + \frac{1}{6} f_y^3 \frac{\partial^2}{(\partial y)^2} f_y + \frac{1}{6} f_x \frac{\partial^2}{(\partial x)^2} f_x - \frac{1}{3} \frac{\partial}{\partial x} f_x \frac{\partial}{\partial y} f_x + \frac{1}{6} f_x \frac{\partial^2}{(\partial y)^2} f_x + \frac{1}{6} f_y \frac{\partial^2}{(\partial x)^2} f_y - \frac{1}{3} \frac{\partial}{\partial x} f_y \frac{\partial}{\partial y} f_y + \frac{1}{6} f_y \frac{\partial^2}{(\partial y)^2} f_y - \frac{1}{12} \frac{\partial^2}{(\partial v)^2} H^0(u, v)
\end{aligned}
$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\begin{aligned}\Psi_2 = {} & \frac{1}{3}\, f_x^2 \frac{\partial}{\partial x} f_x^2 - \frac{1}{3}\, f_y^2 \frac{\partial}{\partial x} f_x^2 - \frac{1}{3}\, f_x^3 \frac{\partial^2}{(\partial x)^2} f_x - \frac{1}{3}\, f_x f_y^2 \frac{\partial^2}{(\partial x)^2} f_x + \frac{1}{3}\, f_x^2 \frac{\partial}{\partial y} f_x^2 - \frac{1}{3}\, f_y^2 \frac{\partial}{\partial y} f_x^2 - \frac{1}{3}\, f_x^3 \frac{\partial^2}{(\partial y)^2} f_x - \frac{1}{3}\, f_x f_y^2 \frac{\partial^2}{(\partial y)^2} f_x + \frac{4}{3}\, f_x f_y \frac{\partial}{\partial x} f_x \frac{\partial}{\partial x} f_y \\ & - \frac{1}{3}\, f_x^2 \frac{\partial}{\partial x} f_y^2 + \frac{1}{3}\, f_y^2 \frac{\partial}{\partial x} f_y^2 - \frac{1}{3}\, f_x^2 f_y \frac{\partial^2}{(\partial x)^2} f_y - \frac{1}{3}\, f_y^3 \frac{\partial^2}{(\partial x)^2} f_y + \frac{4}{3}\, f_x f_y \frac{\partial}{\partial y} f_x \frac{\partial}{\partial y} f_y - \frac{1}{3}\, f_x^2 \frac{\partial}{\partial y} f_y^2 + \frac{1}{3}\, f_y^2 \frac{\partial}{\partial y} f_y^2 - \frac{1}{3}\, f_x^2 f_y \frac{\partial^2}{(\partial y)^2} f_y \\ & - \frac{1}{3}\, f_y^3 \frac{\partial^2}{(\partial y)^2} f_y - \frac{1}{3}\, f_x \frac{\partial^2}{(\partial x)^2} f_x + \frac{2}{3}\, \frac{\partial}{\partial x} f_x \frac{\partial}{\partial y} f_x - \frac{1}{3}\, f_x \frac{\partial^2}{(\partial y)^2} f_x - \frac{1}{3}\, f_y \frac{\partial^2}{(\partial x)^2} f_y + \frac{2}{3}\, \frac{\partial}{\partial x} f_y \frac{\partial}{\partial y} f_y - \frac{1}{3}\, f_y \frac{\partial^2}{(\partial y)^2} f_y + \frac{1}{6}\, \frac{\partial^2}{(\partial v)^2} H^0(u,v)\end{aligned}$$

$$\begin{aligned}\Psi_3 = {} & -\sqrt{2} v f_x^2 \frac{\partial}{\partial x} f_x \frac{\partial^2}{(\partial x)^2} f_x - \sqrt{2} v f_y^2 \frac{\partial}{\partial x} f_x \frac{\partial^2}{(\partial x)^2} f_x + (i-1)\sqrt{2} v f_x^2 \frac{\partial}{\partial x} f_x \frac{\partial^2}{\partial x \partial y} f_x + (i-1)\sqrt{2} v f_y^2 \frac{\partial}{\partial x} f_x \frac{\partial^2}{\partial x \partial y} f_x - \sqrt{2} v f_x^2 \frac{\partial^2}{(\partial x)^2} f_x \frac{\partial}{\partial y} f_x - \sqrt{2} v f_y^2 \frac{\partial^2}{(\partial x)^2} f_x \frac{\partial}{\partial y} f_x \\ & + (i-1)\sqrt{2} v f_x^2 \frac{\partial^2}{\partial x \partial y} f_x \frac{\partial}{\partial y} f_x + (i-1)\sqrt{2} v f_y^2 \frac{\partial^2}{\partial x \partial y} f_x \frac{\partial}{\partial y} f_x + i\sqrt{2} v f_x^2 \frac{\partial}{\partial x} f_x \frac{\partial^2}{(\partial y)^2} f_x + i\sqrt{2} v f_y^2 \frac{\partial}{\partial x} f_x \frac{\partial^2}{(\partial y)^2} f_x + i\sqrt{2} v f_x^2 \frac{\partial}{\partial y} f_x \frac{\partial^2}{(\partial y)^2} f_x + i\sqrt{2} v f_y^2 \frac{\partial}{\partial y} f_x \frac{\partial^2}{(\partial y)^2} f_x \\ & - \sqrt{2} v f_x^2 \frac{\partial}{\partial x} f_y \frac{\partial^2}{(\partial x)^2} f_y - \sqrt{2} v f_y^2 \frac{\partial}{\partial x} f_y \frac{\partial^2}{(\partial x)^2} f_y + (i-1)\sqrt{2} v f_x^2 \frac{\partial}{\partial x} f_y \frac{\partial^2}{\partial x \partial y} f_y + (i-1)\sqrt{2} v f_y^2 \frac{\partial}{\partial x} f_y \frac{\partial^2}{\partial x \partial y} f_y - \sqrt{2} v f_x^2 \frac{\partial^2}{(\partial x)^2} f_y \frac{\partial}{\partial y} f_y - \sqrt{2} v f_y^2 \frac{\partial^2}{(\partial x)^2} f_y \frac{\partial}{\partial y} f_y \\ & + (i-1)\sqrt{2} v f_x^2 \frac{\partial^2}{\partial x \partial y} f_y \frac{\partial}{\partial y} f_y + (i-1)\sqrt{2} v f_y^2 \frac{\partial^2}{\partial x \partial y} f_y \frac{\partial}{\partial y} f_y + i\sqrt{2} v f_x^2 \frac{\partial}{\partial x} f_y \frac{\partial^2}{(\partial y)^2} f_y + i\sqrt{2} v f_y^2 \frac{\partial}{\partial x} f_y \frac{\partial^2}{(\partial y)^2} f_y + i\sqrt{2} v f_x^2 \frac{\partial}{\partial y} f_y \frac{\partial^2}{(\partial y)^2} f_y + i\sqrt{2} v f_y^2 \frac{\partial}{\partial y} f_y \frac{\partial^2}{(\partial y)^2} f_y \\ & - \sqrt{2} v \frac{\partial}{\partial x} f_x \frac{\partial^2}{(\partial x)^2} f_x + (i-1)\sqrt{2} v \frac{\partial}{\partial x} f_x \frac{\partial^2}{\partial x \partial y} f_x - \sqrt{2} v \frac{\partial^2}{(\partial x)^2} f_x \frac{\partial}{\partial y} f_x + (i-1)\sqrt{2} v \frac{\partial^2}{\partial x \partial y} f_x \frac{\partial}{\partial y} f_x + i\sqrt{2} v \frac{\partial}{\partial x} f_x \frac{\partial^2}{(\partial y)^2} f_x + i\sqrt{2} v \frac{\partial}{\partial y} f_x \frac{\partial^2}{(\partial y)^2} f_x \\ & - \sqrt{2} v \frac{\partial}{\partial x} f_y \frac{\partial^2}{(\partial x)^2} f_y + (i-1)\sqrt{2} v \frac{\partial}{\partial x} f_y \frac{\partial^2}{\partial x \partial y} f_y - \sqrt{2} v \frac{\partial^2}{(\partial x)^2} f_y \frac{\partial}{\partial y} f_y + (i-1)\sqrt{2} v \frac{\partial^2}{\partial x \partial y} f_y \frac{\partial}{\partial y} f_y + i\sqrt{2} v \frac{\partial}{\partial x} f_y \frac{\partial^2}{(\partial y)^2} f_y + i\sqrt{2} v \frac{\partial}{\partial y} f_y \frac{\partial^2}{(\partial y)^2} f_y\end{aligned}$$

$$
\begin{aligned}
\Psi_4 = {} & 4\,v^2 f_x^3 \frac{\partial}{\partial x} f_x^2 \frac{\partial^2}{(\partial x)^2} f_x + 4\,v^2 f_x f_y^2 \frac{\partial}{\partial x} f_x^2 \frac{\partial^2}{(\partial x)^2} f_x + v^2 f_x^4 \frac{\partial^2}{(\partial x)^2} f_x^2 + 2\,v^2 f_x^2 f_y^2 \frac{\partial^2}{(\partial x)^2} f_x^2 + v^2 f_y^4 \frac{\partial^2}{(\partial x)^2} f_x^2 + v^2 f_x^4 \frac{\partial}{\partial x} f_x \frac{\partial^3}{(\partial x)^3} f_x + 2\,v^2 f_x^2 f_y^2 \frac{\partial}{\partial x} f_x \frac{\partial^3}{(\partial x)^3} f_x + v^2 f_y^4 \frac{\partial}{\partial x} f_x \frac{\partial^3}{(\partial x)^3} f_x - (2i-1)\,v^2 f_x^4 \frac{\partial}{\partial x} f_x \frac{\partial^3}{(\partial x)^2 \partial y} f_x \\
& - (4i-2)\,v^2 f_x^2 f_y^2 \frac{\partial}{\partial x} f_x \frac{\partial^3}{(\partial x)^2 \partial y} f_x - (2i-1)\,v^2 f_y^4 \frac{\partial}{\partial x} f_x \frac{\partial^3}{(\partial x)^2 \partial y} f_x - (4i-4)\,v^2 f_x^3 \frac{\partial}{\partial x} f_x^2 \frac{\partial^2}{\partial x \partial y} f_x - (4i-4)\,v^2 f_x f_y^2 \frac{\partial}{\partial x} f_x^2 \frac{\partial^2}{\partial x \partial y} f_x - (2i-2)\,v^2 f_x^4 \frac{\partial^2}{(\partial x)^2} f_x \frac{\partial^2}{\partial x \partial y} f_x - (4i-4)\,v^2 f_x^2 f_y^2 \frac{\partial^2}{(\partial x)^2} f_x \frac{\partial^2}{\partial x \partial y} f_x \\
& - (2i-2)\,v^2 f_y^4 \frac{\partial^2}{(\partial x)^2} f_x \frac{\partial^2}{\partial x \partial y} f_x - 2i\,v^2 f_x^4 \frac{\partial^2}{\partial x \partial y} f_x^2 - 4i\,v^2 f_x^2 f_y^2 \frac{\partial^2}{\partial x \partial y} f_x^2 - 2i\,v^2 f_y^4 \frac{\partial^2}{\partial x \partial y} f_x^2 - (2i+1)\,v^2 f_x^4 \frac{\partial}{\partial x} f_x \frac{\partial^3}{\partial x (\partial y)^2} f_x - (4i+2)\,v^2 f_x^2 f_y^2 \frac{\partial}{\partial x} f_x \frac{\partial^3}{\partial x (\partial y)^2} f_x - (2i+1)\,v^2 f_y^4 \frac{\partial}{\partial x} f_x \frac{\partial^3}{\partial x (\partial y)^2} f_x \\
& - (4i-4)\,v^2 f_x^3 \frac{\partial}{\partial x} f_x \frac{\partial^2}{(\partial x)^2} f_x \frac{\partial}{\partial y} f_x - (4i-4)\,v^2 f_x f_y^2 \frac{\partial}{\partial x} f_x \frac{\partial^2}{(\partial x)^2} f_x \frac{\partial}{\partial y} f_x + v^2 f_x^4 \frac{\partial^3}{(\partial x)^3} f_x \frac{\partial}{\partial y} f_x + 2\,v^2 f_x^2 f_y^2 \frac{\partial^3}{(\partial x)^3} f_x \frac{\partial}{\partial y} f_x + v^2 f_y^4 \frac{\partial^3}{(\partial x)^3} f_x \frac{\partial}{\partial y} f_x - (2i-1)\,v^2 f_x^4 \frac{\partial^3}{(\partial x)^2 \partial y} f_x \frac{\partial}{\partial y} f_x - (4i-2)\,v^2 f_x^2 f_y^2 \frac{\partial^3}{(\partial x)^2 \partial y} f_x \frac{\partial}{\partial y} f_x \\
& - (2i-1)\,v^2 f_y^4 \frac{\partial^3}{(\partial x)^2 \partial y} f_x \frac{\partial}{\partial y} f_x - 8i\,v^2 f_x^3 \frac{\partial}{\partial x} f_x \frac{\partial^2}{\partial x \partial y} f_x \frac{\partial}{\partial y} f_x - 8i\,v^2 f_x f_y^2 \frac{\partial}{\partial x} f_x \frac{\partial^2}{\partial x \partial y} f_x \frac{\partial}{\partial y} f_x - (2i+1)\,v^2 f_x^4 \frac{\partial^3}{\partial x (\partial y)^2} f_x \frac{\partial}{\partial y} f_x - (4i+2)\,v^2 f_x^2 f_y^2 \frac{\partial^3}{\partial x (\partial y)^2} f_x \frac{\partial}{\partial y} f_x - (2i+1)\,v^2 f_y^4 \frac{\partial^3}{\partial x (\partial y)^2} f_x \frac{\partial}{\partial y} f_x \\
& - 4i\,v^2 f_x^3 \frac{\partial^2}{(\partial x)^2} f_x \frac{\partial}{\partial y} f_x^2 - 4i\,v^2 f_x f_y^2 \frac{\partial^2}{(\partial x)^2} f_x \frac{\partial}{\partial y} f_x^2 - (4i+4)\,v^2 f_x^3 \frac{\partial^2}{\partial x \partial y} f_x \frac{\partial}{\partial y} f_x^2 - (4i+4)\,v^2 f_x f_y^2 \frac{\partial^2}{\partial x \partial y} f_x \frac{\partial}{\partial y} f_x^2 - 4i\,v^2 f_x^3 \frac{\partial}{\partial x} f_x^2 \frac{\partial^2}{(\partial y)^2} f_x - 4i\,v^2 f_x f_y^2 \frac{\partial}{\partial x} f_x^2 \frac{\partial^2}{(\partial y)^2} f_x - 2i\,v^2 f_x^4 \frac{\partial^2}{(\partial x)^2} f_x \frac{\partial^2}{(\partial y)^2} f_x \\
& - 4i\,v^2 f_x^2 f_y^2 \frac{\partial^2}{(\partial x)^2} f_x \frac{\partial^2}{(\partial y)^2} f_x - 2i\,v^2 f_y^4 \frac{\partial^2}{(\partial x)^2} f_x \frac{\partial^2}{(\partial y)^2} f_x - (2i+2)\,v^2 f_x^4 \frac{\partial^2}{\partial x \partial y} f_x \frac{\partial^2}{(\partial y)^2} f_x - (4i+4)\,v^2 f_x^2 f_y^2 \frac{\partial^2}{\partial x \partial y} f_x \frac{\partial^2}{(\partial y)^2} f_x - (2i+2)\,v^2 f_y^4 \frac{\partial^2}{\partial x \partial y} f_x \frac{\partial^2}{(\partial y)^2} f_x - (4i+4)\,v^2 f_x^3 \frac{\partial}{\partial x} f_x \frac{\partial}{\partial y} f_x \frac{\partial^2}{(\partial y)^2} f_x \\
& - (4i+4)\,v^2 f_x f_y^2 \frac{\partial}{\partial x} f_x \frac{\partial}{\partial y} f_x \frac{\partial^2}{(\partial y)^2} f_x - 4\,v^2 f_x^3 \frac{\partial}{\partial y} f_x^2 \frac{\partial^2}{(\partial y)^2} f_x - 4\,v^2 f_x f_y^2 \frac{\partial}{\partial y} f_x^2 \frac{\partial^2}{(\partial y)^2} f_x - v^2 f_x^4 \frac{\partial^2}{(\partial y)^2} f_x^2 - 2\,v^2 f_x^2 f_y^2 \frac{\partial^2}{(\partial y)^2} f_x^2 - v^2 f_y^4 \frac{\partial^2}{(\partial y)^2} f_x^2 - v^2 f_x^4 \frac{\partial}{\partial x} f_x \frac{\partial^3}{(\partial y)^3} f_x - 2\,v^2 f_x^2 f_y^2 \frac{\partial}{\partial x} f_x \frac{\partial^3}{(\partial y)^3} f_x - v^2 f_y^4 \frac{\partial}{\partial x} f_x \frac{\partial^3}{(\partial y)^3} f_x \\
& - v^2 f_x^4 \frac{\partial}{\partial y} f_x \frac{\partial^3}{(\partial y)^3} f_x - 2\,v^2 f_x^2 f_y^2 \frac{\partial}{\partial y} f_x \frac{\partial^3}{(\partial y)^3} f_x - v^2 f_y^4 \frac{\partial}{\partial y} f_x \frac{\partial^3}{(\partial y)^3} f_x + 4\,v^2 f_x^2 f_y \frac{\partial}{\partial x} f_x \frac{\partial^2}{(\partial x)^2} f_x \frac{\partial}{\partial x} f_y + 4\,v^2 f_y^3 \frac{\partial}{\partial x} f_x \frac{\partial^2}{(\partial x)^2} f_x \frac{\partial}{\partial x} f_y - (4i-4)\,v^2 f_x^2 f_y \frac{\partial}{\partial x} f_x \frac{\partial^2}{\partial x \partial y} f_x \frac{\partial}{\partial x} f_y - (4i-4)\,v^2 f_y^3 \frac{\partial}{\partial x} f_x \frac{\partial^2}{\partial x \partial y} f_x \frac{\partial}{\partial x} f_y \\
& + 4\,v^2 f_x^2 f_y \frac{\partial^2}{(\partial x)^2} f_x \frac{\partial}{\partial y} f_x \frac{\partial}{\partial x} f_y + 4\,v^2 f_y^3 \frac{\partial^2}{(\partial x)^2} f_x \frac{\partial}{\partial y} f_x \frac{\partial}{\partial x} f_y - (4i-4)\,v^2 f_x^2 f_y \frac{\partial^2}{\partial x \partial y} f_x \frac{\partial}{\partial y} f_x \frac{\partial}{\partial x} f_y - (4i-4)\,v^2 f_y^3 \frac{\partial^2}{\partial x \partial y} f_x \frac{\partial}{\partial y} f_x \frac{\partial}{\partial x} f_y - 4i\,v^2 f_x^2 f_y \frac{\partial}{\partial x} f_x \frac{\partial^2}{(\partial y)^2} f_x \frac{\partial}{\partial x} f_y - 4i\,v^2 f_y^3 \frac{\partial}{\partial x} f_x \frac{\partial^2}{(\partial y)^2} f_x \frac{\partial}{\partial x} f_y \\
& - 4i\,v^2 f_x^2 f_y \frac{\partial}{\partial y} f_x \frac{\partial^2}{(\partial y)^2} f_x \frac{\partial}{\partial x} f_y - 4i\,v^2 f_y^3 \frac{\partial}{\partial y} f_x \frac{\partial^2}{(\partial y)^2} f_x \frac{\partial}{\partial x} f_y + 4\,v^2 f_x^3 \frac{\partial}{\partial x} f_x \frac{\partial}{\partial x} f_y \frac{\partial^2}{(\partial x)^2} f_y + 4\,v^2 f_x f_y^2 \frac{\partial}{\partial x} f_x \frac{\partial}{\partial x} f_y \frac{\partial^2}{(\partial x)^2} f_y - 4i\,v^2 f_x^3 \frac{\partial}{\partial y} f_x \frac{\partial}{\partial x} f_y \frac{\partial^2}{(\partial x)^2} f_y - 4i\,v^2 f_x f_y^2 \frac{\partial}{\partial y} f_x \frac{\partial}{\partial x} f_y \frac{\partial^2}{(\partial x)^2} f_y \\
& + 4\,v^2 f_x^2 f_y \frac{\partial}{\partial x} f_y^2 \frac{\partial^2}{(\partial x)^2} f_y + 4\,v^2 f_y^3 \frac{\partial}{\partial x} f_y^2 \frac{\partial^2}{(\partial x)^2} f_y + v^2 f_x^4 \frac{\partial^2}{(\partial x)^2} f_y^2 + 2\,v^2 f_x^2 f_y^2 \frac{\partial^2}{(\partial x)^2} f_y^2 + v^2 f_y^4 \frac{\partial^2}{(\partial x)^2} f_y^2 + v^2 f_x^4 \frac{\partial}{\partial x} f_y \frac{\partial^3}{(\partial x)^3} f_y + 2\,v^2 f_x^2 f_y^2 \frac{\partial}{\partial x} f_y \frac{\partial^3}{(\partial x)^3} f_y + v^2 f_y^4 \frac{\partial}{\partial x} f_y \frac{\partial^3}{(\partial x)^3} f_y - (2i-1)\,v^2 f_x^4 \frac{\partial}{\partial x} f_y \frac{\partial^3}{(\partial x)^2 \partial y} f_y \\
& - (4i-2)\,v^2 f_x^2 f_y^2 \frac{\partial}{\partial x} f_y \frac{\partial^3}{(\partial x)^2 \partial y} f_y - (2i-1)\,v^2 f_y^4 \frac{\partial}{\partial x} f_y \frac{\partial^3}{(\partial x)^2 \partial y} f_y - (4i-4)\,v^2 f_x^3 \frac{\partial}{\partial x} f_x \frac{\partial}{\partial x} f_y \frac{\partial^2}{\partial x \partial y} f_y - (4i-4)\,v^2 f_x f_y^2 \frac{\partial}{\partial x} f_x \frac{\partial}{\partial x} f_y \frac{\partial^2}{\partial x \partial y} f_y - (4i+4)\,v^2 f_x^3 \frac{\partial}{\partial y} f_x \frac{\partial}{\partial x} f_y \frac{\partial^2}{\partial x \partial y} f_y - (4i+4)\,v^2 f_x f_y^2 \frac{\partial}{\partial y} f_x \frac{\partial}{\partial x} f_y \frac{\partial^2}{\partial x \partial y} f_y \\
& - (4i-4)\,v^2 f_x^2 f_y \frac{\partial}{\partial x} f_y^2 \frac{\partial^2}{\partial x \partial y} f_y - (4i-4)\,v^2 f_y^3 \frac{\partial}{\partial x} f_y^2 \frac{\partial^2}{\partial x \partial y} f_y - (2i-2)\,v^2 f_x^4 \frac{\partial^2}{(\partial x)^2} f_y \frac{\partial^2}{\partial x \partial y} f_y - (4i-4)\,v^2 f_x^2 f_y^2 \frac{\partial^2}{(\partial x)^2} f_y \frac{\partial^2}{\partial x \partial y} f_y - (2i-2)\,v^2 f_y^4 \frac{\partial^2}{(\partial x)^2} f_y \frac{\partial^2}{\partial x \partial y} f_y - 2i\,v^2 f_x^4 \frac{\partial^2}{\partial x \partial y} f_y^2 - 4i\,v^2 f_x^2 f_y^2 \frac{\partial^2}{\partial x \partial y} f_y^2 \\
& - 2i\,v^2 f_y^4 \frac{\partial^2}{\partial x \partial y} f_y^2 - (2i+1)\,v^2 f_x^4 \frac{\partial}{\partial x} f_y \frac{\partial^3}{\partial x (\partial y)^2} f_y - (4i+2)\,v^2 f_x^2 f_y^2 \frac{\partial}{\partial x} f_y \frac{\partial^3}{\partial x (\partial y)^2} f_y - (2i+1)\,v^2 f_y^4 \frac{\partial}{\partial x} f_y \frac{\partial^3}{\partial x (\partial y)^2} f_y - 4i\,v^2 f_x^2 f_y \frac{\partial}{\partial x} f_x \frac{\partial^2}{(\partial x)^2} f_x \frac{\partial}{\partial y} f_y - 4i\,v^2 f_y^3 \frac{\partial}{\partial x} f_x \frac{\partial^2}{(\partial x)^2} f_x \frac{\partial}{\partial y} f_y \\
& - (4i+4)\,v^2 f_x^2 f_y \frac{\partial}{\partial x} f_x \frac{\partial^2}{\partial x \partial y} f_x \frac{\partial}{\partial y} f_y - (4i+4)\,v^2 f_y^3 \frac{\partial}{\partial x} f_x \frac{\partial^2}{\partial x \partial y} f_x \frac{\partial}{\partial y} f_y - 4i\,v^2 f_x^2 f_y \frac{\partial^2}{(\partial x)^2} f_x \frac{\partial}{\partial y} f_x \frac{\partial}{\partial y} f_y - 4i\,v^2 f_y^3 \frac{\partial^2}{(\partial x)^2} f_x \frac{\partial}{\partial y} f_x \frac{\partial}{\partial y} f_y - (4i+4)\,v^2 f_x^2 f_y \frac{\partial^2}{\partial x \partial y} f_x \frac{\partial}{\partial y} f_x \frac{\partial}{\partial y} f_y - (4i+4)\,v^2 f_y^3 \frac{\partial^2}{\partial x \partial y} f_x \frac{\partial}{\partial y} f_x \frac{\partial}{\partial y} f_y \\
& - 4\,v^2 f_x^2 f_y \frac{\partial}{\partial x} f_x \frac{\partial^2}{(\partial y)^2} f_x \frac{\partial}{\partial y} f_y - 4\,v^2 f_y^3 \frac{\partial}{\partial x} f_x \frac{\partial^2}{(\partial y)^2} f_x \frac{\partial}{\partial y} f_y - 4\,v^2 f_x^2 f_y \frac{\partial}{\partial y} f_x \frac{\partial^2}{(\partial y)^2} f_x \frac{\partial}{\partial y} f_y - 4\,v^2 f_y^3 \frac{\partial}{\partial y} f_x \frac{\partial^2}{(\partial y)^2} f_x \frac{\partial}{\partial y} f_y + 4\,v^2 f_x^3 \frac{\partial}{\partial x} f_x \frac{\partial^2}{(\partial x)^2} f_y \frac{\partial}{\partial y} f_y + 4\,v^2 f_x f_y^2 \frac{\partial}{\partial x} f_x \frac{\partial^2}{(\partial x)^2} f_y \frac{\partial}{\partial y} f_y - 4i\,v^2 f_x^3 \frac{\partial}{\partial y} f_x \frac{\partial^2}{(\partial x)^2} f_y \frac{\partial}{\partial y} f_y \\
& - 4i\,v^2 f_x f_y^2 \frac{\partial}{\partial y} f_x \frac{\partial^2}{(\partial x)^2} f_y \frac{\partial}{\partial y} f_y - (4i-4)\,v^2 f_x^2 f_y \frac{\partial}{\partial x} f_y \frac{\partial^2}{(\partial x)^2} f_y \frac{\partial}{\partial y} f_y - (4i-4)\,v^2 f_y^3 \frac{\partial}{\partial x} f_y \frac{\partial^2}{(\partial x)^2} f_y \frac{\partial}{\partial y} f_y + v^2 f_x^4 \frac{\partial^3}{(\partial x)^3} f_y \frac{\partial}{\partial y} f_y + 2\,v^2 f_x^2 f_y^2 \frac{\partial^3}{(\partial x)^3} f_y \frac{\partial}{\partial y} f_y + v^2 f_y^4 \frac{\partial^3}{(\partial x)^3} f_y \frac{\partial}{\partial y} f_y - (2i-1)\,v^2 f_x^4 \frac{\partial^3}{(\partial x)^2 \partial y} f_y \frac{\partial}{\partial y} f_y \\
& - (4i-2)\,v^2 f_x^2 f_y^2 \frac{\partial^3}{(\partial x)^2 \partial y} f_y \frac{\partial}{\partial y} f_y - (2i-1)\,v^2 f_y^4 \frac{\partial^3}{(\partial x)^2 \partial y} f_y \frac{\partial}{\partial y} f_y - (4i-4)\,v^2 f_x^3 \frac{\partial}{\partial x} f_x \frac{\partial^2}{\partial x \partial y} f_y \frac{\partial}{\partial y} f_y - (4i-4)\,v^2 f_x f_y^2 \frac{\partial}{\partial x} f_x \frac{\partial^2}{\partial x \partial y} f_y \frac{\partial}{\partial y} f_y - (4i+4)\,v^2 f_x^3 \frac{\partial}{\partial y} f_x \frac{\partial^2}{\partial x \partial y} f_y \frac{\partial}{\partial y} f_y - (4i+4)\,v^2 f_x f_y^2 \frac{\partial}{\partial y} f_x \frac{\partial^2}{\partial x \partial y} f_y \frac{\partial}{\partial y} f_y \\
& - 8i\,v^2 f_x^2 f_y \frac{\partial}{\partial x} f_y \frac{\partial^2}{\partial x \partial y} f_y \frac{\partial}{\partial y} f_y - 8i\,v^2 f_y^3 \frac{\partial}{\partial x} f_y \frac{\partial^2}{\partial x \partial y} f_y \frac{\partial}{\partial y} f_y - (2i+1)\,v^2 f_x^4 \frac{\partial^3}{\partial x (\partial y)^2} f_y \frac{\partial}{\partial y} f_y - (4i+2)\,v^2 f_x^2 f_y^2 \frac{\partial^3}{\partial x (\partial y)^2} f_y \frac{\partial}{\partial y} f_y - (2i+1)\,v^2 f_y^4 \frac{\partial^3}{\partial x (\partial y)^2} f_y \frac{\partial}{\partial y} f_y - 4i\,v^2 f_x^2 f_y \frac{\partial^2}{(\partial x)^2} f_y \frac{\partial}{\partial y} f_y^2 - 4i\,v^2 f_y^3 \frac{\partial^2}{(\partial x)^2} f_y \frac{\partial}{\partial y} f_y^2 \\
& - (4i+4)\,v^2 f_x^2 f_y \frac{\partial^2}{\partial x \partial y} f_y \frac{\partial}{\partial y} f_y^2 - (4i+4)\,v^2 f_y^3 \frac{\partial^2}{\partial x \partial y} f_y \frac{\partial}{\partial y} f_y^2 - 4i\,v^2 f_x^3 \frac{\partial}{\partial x} f_x \frac{\partial}{\partial x} f_y \frac{\partial^2}{(\partial y)^2} f_y - 4i\,v^2 f_x f_y^2 \frac{\partial}{\partial x} f_x \frac{\partial}{\partial x} f_y \frac{\partial^2}{(\partial y)^2} f_y - 4\,v^2 f_x^3 \frac{\partial}{\partial y} f_x \frac{\partial}{\partial x} f_y \frac{\partial^2}{(\partial y)^2} f_y - 4\,v^2 f_x f_y^2 \frac{\partial}{\partial y} f_x \frac{\partial}{\partial x} f_y \frac{\partial^2}{(\partial y)^2} f_y - 4i\,v^2 f_x^2 f_y \frac{\partial}{\partial x} f_y^2 \frac{\partial^2}{(\partial y)^2} f_y
\end{aligned}
$$

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\begin{aligned}
&-4i\,v^2 f_y^3 \frac{\partial}{\partial x} f_y^2 \frac{\partial^2}{(\partial y)^2} f_y - 2i\,v^2 f_x^4 \frac{\partial^2}{(\partial x)^2} f_y \frac{\partial^2}{(\partial y)^2} f_y - 4i\,v^2 f_x^2 f_y^2 \frac{\partial^2}{(\partial x)^2} f_y \frac{\partial^2}{(\partial y)^2} f_y - 2i\,v^2 f_y^4 \frac{\partial^2}{(\partial x)^2} f_y \frac{\partial^2}{(\partial y)^2} f_y - (2i+2)\,v^2 f_x^4 \frac{\partial^2}{\partial x \partial y} f_y \frac{\partial^2}{(\partial y)^2} f_y - (4i+4)\,v^2 f_x^2 f_y^2 \frac{\partial^2}{\partial x \partial y} f_y \frac{\partial^2}{(\partial y)^2} f_y \\
&- (2i+2)\,v^2 f_y^4 \frac{\partial^2}{\partial x \partial y} f_y \frac{\partial^2}{(\partial y)^2} f_y - 4i\,v^2 f_x^3 \frac{\partial}{\partial x} f_x \frac{\partial}{\partial y} f_y \frac{\partial^2}{(\partial y)^2} f_y - 4i\,v^2 f_x f_y^2 \frac{\partial}{\partial x} f_x \frac{\partial}{\partial y} f_y \frac{\partial^2}{(\partial y)^2} f_y - 4\,v^2 f_x^3 \frac{\partial}{\partial y} f_x \frac{\partial}{\partial y} f_y \frac{\partial^2}{(\partial y)^2} f_y - 4\,v^2 f_x f_y^2 \frac{\partial}{\partial y} f_x \frac{\partial}{\partial y} f_y \frac{\partial^2}{(\partial y)^2} f_y \\
&- (4i+4)\,v^2 f_x^2 f_y \frac{\partial}{\partial x} f_y \frac{\partial}{\partial y} f_y \frac{\partial^2}{(\partial y)^2} f_y - (4i+4)\,v^2 f_y^3 \frac{\partial}{\partial x} f_y \frac{\partial}{\partial y} f_y \frac{\partial^2}{(\partial y)^2} f_y - 4\,v^2 f_x^2 f_y \frac{\partial}{\partial y} f_y^2 \frac{\partial^2}{(\partial y)^2} f_y - 4\,v^2 f_y^3 \frac{\partial}{\partial y} f_y^2 \frac{\partial^2}{(\partial y)^2} f_y - v^2 f_x^4 \frac{\partial^2}{(\partial y)^2} f_y^2 - 2\,v^2 f_x^2 f_y^2 \frac{\partial^2}{(\partial y)^2} f_y^2 - v^2 f_y^4 \frac{\partial^2}{(\partial y)^2} f_y^2 \\
&- v^2 f_x^4 \frac{\partial}{\partial x} f_y \frac{\partial^3}{(\partial y)^3} f_y - 2\,v^2 f_x^2 f_y^2 \frac{\partial}{\partial x} f_y \frac{\partial^3}{(\partial y)^3} f_y - v^2 f_y^4 \frac{\partial}{\partial x} f_y \frac{\partial^3}{(\partial y)^3} f_y - v^2 f_x^4 \frac{\partial}{\partial y} f_y \frac{\partial^3}{(\partial y)^3} f_y - 2\,v^2 f_x^2 f_y^2 \frac{\partial}{\partial y} f_y \frac{\partial^3}{(\partial y)^3} f_y - v^2 f_y^4 \frac{\partial}{\partial y} f_y \frac{\partial^3}{(\partial y)^3} f_y + 4\,v^2 f_x \frac{\partial}{\partial x} f_x^2 \frac{\partial^2}{(\partial x)^2} f_x + 2\,v^2 f_x^2 \frac{\partial^2}{(\partial x)^2} f_x^2 \\
&+ 2\,v^2 f_y^2 \frac{\partial^2}{(\partial x)^2} f_x^2 + 2\,v^2 f_x^2 \frac{\partial}{\partial x} f_x \frac{\partial^3}{(\partial x)^3} f_x + 2\,v^2 f_y^2 \frac{\partial}{\partial x} f_x \frac{\partial^3}{(\partial x)^3} f_x - (4i-2)\,v^2 f_x^2 \frac{\partial}{\partial x} f_x \frac{\partial^3}{(\partial x)^2 \partial y} f_x - (4i-2)\,v^2 f_y^2 \frac{\partial}{\partial x} f_x \frac{\partial^3}{(\partial x)^2 \partial y} f_x - (4i-4)\,v^2 f_x \frac{\partial}{\partial x} f_x^2 \frac{\partial^2}{\partial x \partial y} f_x - (4i-4)\,v^2 f_x^2 \frac{\partial^2}{(\partial x)^2} f_x \frac{\partial^2}{\partial x \partial y} f_x \\
&- (4i-4)\,v^2 f_y^2 \frac{\partial^2}{(\partial x)^2} f_x \frac{\partial^2}{\partial x \partial y} f_x - 4i\,v^2 f_x^2 \frac{\partial^2}{\partial x \partial y} f_x^2 - 4i\,v^2 f_y^2 \frac{\partial^2}{\partial x \partial y} f_x^2 - (4i+2)\,v^2 f_x^2 \frac{\partial}{\partial x} f_x \frac{\partial^3}{\partial x (\partial y)^2} f_x - (4i+2)\,v^2 f_y^2 \frac{\partial}{\partial x} f_x \frac{\partial^3}{\partial x (\partial y)^2} f_x - (4i-4)\,v^2 f_x \frac{\partial}{\partial x} f_x \frac{\partial^2}{(\partial x)^2} f_x \frac{\partial}{\partial y} f_x + 2\,v^2 f_x^2 \frac{\partial^3}{(\partial x)^3} f_x \frac{\partial}{\partial y} f_x \\
&+ 2\,v^2 f_y^2 \frac{\partial^3}{(\partial x)^3} f_x \frac{\partial}{\partial y} f_x - (4i-2)\,v^2 f_x^2 \frac{\partial^3}{(\partial x)^2 \partial y} f_x \frac{\partial}{\partial y} f_x - (4i-2)\,v^2 f_y^2 \frac{\partial^3}{(\partial x)^2 \partial y} f_x \frac{\partial}{\partial y} f_x - 8i\,v^2 f_x \frac{\partial}{\partial x} f_x \frac{\partial^2}{\partial x \partial y} f_x \frac{\partial}{\partial y} f_x - (4i+2)\,v^2 f_x^2 \frac{\partial^3}{\partial x (\partial y)^2} f_x \frac{\partial}{\partial y} f_x - (4i+2)\,v^2 f_y^2 \frac{\partial^3}{\partial x (\partial y)^2} f_x \frac{\partial}{\partial y} f_x \\
&- 4i\,v^2 f_x \frac{\partial^2}{(\partial x)^2} f_x \frac{\partial}{\partial y} f_x^2 - (4i+4)\,v^2 f_x \frac{\partial^2}{\partial x \partial y} f_x \frac{\partial}{\partial y} f_x^2 - 4i\,v^2 f_x \frac{\partial}{\partial x} f_x^2 \frac{\partial^2}{(\partial y)^2} f_x - 4i\,v^2 f_x^2 \frac{\partial^2}{(\partial x)^2} f_x \frac{\partial^2}{(\partial y)^2} f_x - 4i\,v^2 f_y^2 \frac{\partial^2}{(\partial x)^2} f_x \frac{\partial^2}{(\partial y)^2} f_x - (4i+4)\,v^2 f_x^2 \frac{\partial^2}{\partial x \partial y} f_x \frac{\partial^2}{(\partial y)^2} f_x \\
&- (4i+4)\,v^2 f_y^2 \frac{\partial^2}{\partial x \partial y} f_x \frac{\partial^2}{(\partial y)^2} f_x - (4i+4)\,v^2 f_x \frac{\partial}{\partial x} f_x \frac{\partial}{\partial y} f_x \frac{\partial^2}{(\partial y)^2} f_x - 4\,v^2 f_x \frac{\partial}{\partial y} f_x^2 \frac{\partial^2}{(\partial y)^2} f_x - 2\,v^2 f_x^2 \frac{\partial^2}{(\partial y)^2} f_x^2 - 2\,v^2 f_y^2 \frac{\partial^2}{(\partial y)^2} f_x^2 - 2\,v^2 f_x^2 \frac{\partial}{\partial x} f_x \frac{\partial^3}{(\partial y)^3} f_x - 2\,v^2 f_y^2 \frac{\partial}{\partial x} f_x \frac{\partial^3}{(\partial y)^3} f_x \\
&- 2\,v^2 f_x^2 \frac{\partial}{\partial y} f_x \frac{\partial^3}{(\partial y)^3} f_x - 2\,v^2 f_y^2 \frac{\partial}{\partial y} f_x \frac{\partial^3}{(\partial y)^3} f_x + 4\,v^2 f_y \frac{\partial}{\partial x} f_x \frac{\partial^2}{(\partial x)^2} f_x \frac{\partial}{\partial x} f_y - (4i-4)\,v^2 f_y \frac{\partial}{\partial x} f_x \frac{\partial^2}{\partial x \partial y} f_x \frac{\partial}{\partial x} f_y + 4\,v^2 f_y \frac{\partial^2}{(\partial x)^2} f_x \frac{\partial}{\partial y} f_x \frac{\partial}{\partial x} f_y - (4i-4)\,v^2 f_y \frac{\partial^2}{\partial x \partial y} f_x \frac{\partial}{\partial y} f_x \frac{\partial}{\partial x} f_y \\
&- 4i\,v^2 f_y \frac{\partial}{\partial x} f_x \frac{\partial^2}{(\partial y)^2} f_x \frac{\partial}{\partial x} f_y - 4i\,v^2 f_y \frac{\partial}{\partial y} f_x \frac{\partial^2}{(\partial y)^2} f_x \frac{\partial}{\partial x} f_y + 4\,v^2 f_x \frac{\partial}{\partial x} f_x \frac{\partial}{\partial x} f_y \frac{\partial^2}{(\partial x)^2} f_y - 4i\,v^2 f_x \frac{\partial}{\partial y} f_x \frac{\partial}{\partial x} f_y \frac{\partial^2}{(\partial x)^2} f_y + 4\,v^2 f_y \frac{\partial}{\partial x} f_y^2 \frac{\partial^2}{(\partial x)^2} f_y + 2\,v^2 f_x^2 \frac{\partial^2}{(\partial x)^2} f_y^2 + 2\,v^2 f_y^2 \frac{\partial^2}{(\partial x)^2} f_y^2 \\
&+ 2\,v^2 f_x^2 \frac{\partial}{\partial x} f_y \frac{\partial^3}{(\partial x)^3} f_y + 2\,v^2 f_y^2 \frac{\partial}{\partial x} f_y \frac{\partial^3}{(\partial x)^3} f_y - (4i-2)\,v^2 f_x^2 \frac{\partial}{\partial x} f_y \frac{\partial^3}{(\partial x)^2 \partial y} f_y - (4i-2)\,v^2 f_y^2 \frac{\partial}{\partial x} f_y \frac{\partial^3}{(\partial x)^2 \partial y} f_y - (4i-4)\,v^2 f_x \frac{\partial}{\partial x} f_x \frac{\partial}{\partial x} f_y \frac{\partial^2}{\partial x \partial y} f_y - (4i+4)\,v^2 f_x \frac{\partial}{\partial y} f_x \frac{\partial}{\partial x} f_y \frac{\partial^2}{\partial x \partial y} f_y \\
&- (4i-4)\,v^2 f_y \frac{\partial}{\partial x} f_y^2 \frac{\partial^2}{\partial x \partial y} f_y - (4i-4)\,v^2 f_x^2 \frac{\partial^2}{(\partial x)^2} f_y \frac{\partial^2}{\partial x \partial y} f_y - (4i-4)\,v^2 f_y^2 \frac{\partial^2}{(\partial x)^2} f_y \frac{\partial^2}{\partial x \partial y} f_y - 4i\,v^2 f_x^2 \frac{\partial^2}{\partial x \partial y} f_y^2 - 4i\,v^2 f_y^2 \frac{\partial^2}{\partial x \partial y} f_y^2 - (4i+2)\,v^2 f_x^2 \frac{\partial}{\partial x} f_y \frac{\partial^3}{\partial x (\partial y)^2} f_y \\
&- (4i+2)\,v^2 f_y^2 \frac{\partial}{\partial x} f_y \frac{\partial^3}{\partial x (\partial y)^2} f_y - 4i\,v^2 f_y \frac{\partial}{\partial x} f_x \frac{\partial^2}{(\partial x)^2} f_x \frac{\partial}{\partial y} f_y - (4i+4)\,v^2 f_y \frac{\partial}{\partial x} f_x \frac{\partial^2}{\partial x \partial y} f_x \frac{\partial}{\partial y} f_y - 4i\,v^2 f_y \frac{\partial^2}{(\partial x)^2} f_x \frac{\partial}{\partial y} f_x \frac{\partial}{\partial y} f_y - (4i+4)\,v^2 f_y \frac{\partial^2}{\partial x \partial y} f_x \frac{\partial}{\partial y} f_x \frac{\partial}{\partial y} f_y
\end{aligned}
$$

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\begin{aligned}
&-4\,v^2 f_y \frac{\partial}{\partial x} f_x \frac{\partial^2}{(\partial y)^2} f_x \frac{\partial}{\partial y} f_y - 4\,v^2 f_y \frac{\partial}{\partial y} f_x \frac{\partial^2}{(\partial y)^2} f_x \frac{\partial}{\partial y} f_y + 4\,v^2 f_x \frac{\partial}{\partial x} f_x \frac{\partial^2}{(\partial x)^2} f_y \frac{\partial}{\partial y} f_y - 4i\,v^2 f_x \frac{\partial}{\partial y} f_x \frac{\partial^2}{(\partial x)^2} f_y \frac{\partial}{\partial y} f_y - (4i-4)\,v^2 f_y \frac{\partial}{\partial x} f_y \frac{\partial^2}{(\partial x)^2} f_y \frac{\partial}{\partial y} f_y + 2\,v^2 f_x^2 \frac{\partial^3}{(\partial x)^3} f_y \frac{\partial}{\partial y} f_y \\
&+ 2\,v^2 f_y^2 \frac{\partial^3}{(\partial x)^3} f_y \frac{\partial}{\partial y} f_y - (4i-2)\,v^2 f_x^2 \frac{\partial^3}{(\partial x)^2 \partial y} f_y \frac{\partial}{\partial y} f_y - (4i-2)\,v^2 f_y^2 \frac{\partial^3}{(\partial x)^2 \partial y} f_y \frac{\partial}{\partial y} f_y - (4i-4)\,v^2 f_x \frac{\partial}{\partial x} f_x \frac{\partial^2}{\partial x \partial y} f_y \frac{\partial}{\partial y} f_y - (4i+4)\,v^2 f_x \frac{\partial}{\partial y} f_x \frac{\partial^2}{\partial x \partial y} f_y \frac{\partial}{\partial y} f_y - 8i\,v^2 f_y \frac{\partial}{\partial x} f_y \frac{\partial^2}{\partial x \partial y} f_y \frac{\partial}{\partial y} f_y \\
&- (4i+2)\,v^2 f_x^2 \frac{\partial^3}{\partial x (\partial y)^2} f_y \frac{\partial}{\partial y} f_y - (4i+2)\,v^2 f_y^2 \frac{\partial^3}{\partial x (\partial y)^2} f_y \frac{\partial}{\partial y} f_y - 4i\,v^2 f_y \frac{\partial^2}{(\partial x)^2} f_y \frac{\partial}{\partial y} f_y^2 - (4i+4)\,v^2 f_y \frac{\partial^2}{\partial x \partial y} f_y \frac{\partial}{\partial y} f_y^2 - 4i\,v^2 f_x \frac{\partial}{\partial x} f_x \frac{\partial}{\partial x} f_y \frac{\partial^2}{(\partial y)^2} f_y - 4\,v^2 f_x \frac{\partial}{\partial y} f_x \frac{\partial}{\partial x} f_y \frac{\partial^2}{(\partial y)^2} f_y \\
&- 4i\,v^2 f_y \frac{\partial}{\partial x} f_y^2 \frac{\partial^2}{(\partial y)^2} f_y - 4i\,v^2 f_x^2 \frac{\partial^2}{(\partial x)^2} f_y \frac{\partial^2}{(\partial y)^2} f_y - 4i\,v^2 f_y^2 \frac{\partial^2}{(\partial x)^2} f_y \frac{\partial^2}{(\partial y)^2} f_y - (4i+4)\,v^2 f_x^2 \frac{\partial^2}{\partial x \partial y} f_y \frac{\partial^2}{(\partial y)^2} f_y - (4i+4)\,v^2 f_y^2 \frac{\partial^2}{\partial x \partial y} f_y \frac{\partial^2}{(\partial y)^2} f_y - 4i\,v^2 f_x \frac{\partial}{\partial x} f_x \frac{\partial}{\partial y} f_y \frac{\partial^2}{(\partial y)^2} f_y \\
&- 4\,v^2 f_x \frac{\partial}{\partial y} f_x \frac{\partial}{\partial y} f_y \frac{\partial^2}{(\partial y)^2} f_y - (4i+4)\,v^2 f_y \frac{\partial}{\partial x} f_y \frac{\partial}{\partial y} f_y \frac{\partial^2}{(\partial y)^2} f_y - 4\,v^2 f_y \frac{\partial}{\partial y} f_y^2 \frac{\partial^2}{(\partial y)^2} f_y - 2\,v^2 f_x^2 \frac{\partial^2}{(\partial y)^2} f_y^2 - 2\,v^2 f_y^2 \frac{\partial^2}{(\partial y)^2} f_y^2 - 2\,v^2 f_x^2 \frac{\partial}{\partial x} f_y \frac{\partial^3}{(\partial y)^3} f_y - 2\,v^2 f_y^2 \frac{\partial}{\partial x} f_y \frac{\partial^3}{(\partial y)^3} f_y \\
&- 2\,v^2 f_x^2 \frac{\partial}{\partial y} f_y \frac{\partial^3}{(\partial y)^3} f_y - 2\,v^2 f_y^2 \frac{\partial}{\partial y} f_y \frac{\partial^3}{(\partial y)^3} f_y + v^2 \frac{\partial^2}{(\partial x)^2} f_x^2 + v^2 \frac{\partial}{\partial x} f_x \frac{\partial^3}{(\partial x)^3} f_x - (2i-1)\,v^2 \frac{\partial}{\partial x} f_x \frac{\partial^3}{(\partial x)^2 \partial y} f_x - (2i-2)\,v^2 \frac{\partial^2}{(\partial x)^2} f_x \frac{\partial^2}{\partial x \partial y} f_x - 2i\,v^2 \frac{\partial^2}{\partial x \partial y} f_x^2 - (2i+1)\,v^2 \frac{\partial}{\partial x} f_x \frac{\partial^3}{\partial x (\partial y)^2} f_x \\
&+ v^2 \frac{\partial^3}{(\partial x)^3} f_x \frac{\partial}{\partial y} f_x - (2i-1)\,v^2 \frac{\partial^3}{(\partial x)^2 \partial y} f_x \frac{\partial}{\partial y} f_x - (2i+1)\,v^2 \frac{\partial^3}{\partial x (\partial y)^2} f_x \frac{\partial}{\partial y} f_x - 2i\,v^2 \frac{\partial^2}{(\partial x)^2} f_x \frac{\partial^2}{(\partial y)^2} f_x - (2i+2)\,v^2 \frac{\partial^2}{\partial x \partial y} f_x \frac{\partial^2}{(\partial y)^2} f_x - v^2 \frac{\partial^2}{(\partial y)^2} f_x^2 - v^2 \frac{\partial}{\partial x} f_x \frac{\partial^3}{(\partial y)^3} f_x \\
&- v^2 \frac{\partial}{\partial y} f_x \frac{\partial^3}{(\partial y)^3} f_x + v^2 \frac{\partial^2}{(\partial x)^2} f_y^2 + v^2 \frac{\partial}{\partial x} f_y \frac{\partial^3}{(\partial x)^3} f_y - (2i-1)\,v^2 \frac{\partial}{\partial x} f_y \frac{\partial^3}{(\partial x)^2 \partial y} f_y - (2i-2)\,v^2 \frac{\partial^2}{(\partial x)^2} f_y \frac{\partial^2}{\partial x \partial y} f_y - 2i\,v^2 \frac{\partial^2}{\partial x \partial y} f_y^2 - (2i+1)\,v^2 \frac{\partial}{\partial x} f_y \frac{\partial^3}{\partial x (\partial y)^2} f_y + v^2 \frac{\partial^3}{(\partial x)^3} f_y \frac{\partial}{\partial y} f_y \\
&- (2i-1)\,v^2 \frac{\partial^3}{(\partial x)^2 \partial y} f_y \frac{\partial}{\partial y} f_y - (2i+1)\,v^2 \frac{\partial^3}{\partial x (\partial y)^2} f_y \frac{\partial}{\partial y} f_y - 2i\,v^2 \frac{\partial^2}{(\partial x)^2} f_y \frac{\partial^2}{(\partial y)^2} f_y - (2i+2)\,v^2 \frac{\partial^2}{\partial x \partial y} f_y \frac{\partial^2}{(\partial y)^2} f_y - v^2 \frac{\partial^2}{(\partial y)^2} f_y^2 - v^2 \frac{\partial}{\partial x} f_y \frac{\partial^3}{(\partial y)^3} f_y - v^2 \frac{\partial}{\partial y} f_y \frac{\partial^3}{(\partial y)^3} f_y
\end{aligned}
$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "II"